PROJECT PLANNING & EXECUTION ROADMAP

AI Weather Wise

Team Members: Deepika.K, Hemavathy.R, Akash.R **Domain:** AI-Powered Weather Intelligence & Advisory

PHASE PLAN

Phase 1: Concept & Target Audience
Identify real-time weather pain points, outdoor activity planning needs, and AI weather advisor concept.

Phase 2: Requirements Specification
Define functional/non-functional requirements, meteorology data specs, hardware, and security needs.

Phase 3: Architecture & System Design
Design multi-tier architecture, ER/MongoDB schema, OpenWeather/NOAA integrations, and Gemini AI flow.

Phase 4: Planning & Test Strategies
Plan backend modules, API testing strategies, release milestones, and project submission criteria.

Phase 5: Implementation & Integration
Implement Node.js/Express APIs, MongoDB weather models, JWT security, and Gemini advisory engine.

Phase 6: Verification & QA Testing
Test weather telemetry pipelines, authentication, location caching, rate limiting, and AI query response.

Phase 7: Technical Documentation
Finalize API documentation, system architecture specs, user guide, and presentation deck.

Phase 8: Live Demonstration
Record and prepare the live project demonstration highlighting real-time AI weather advisories.

WORK BREAKDOWN

- Authentication & User Profile Security
- Location Preferences & Weather Caching Engine
- Live Weather Data Ingestion & Radar Integration
- Gemini AI Weather Advisory & Alert Summary Engine
- Request Validation & Error Handling Middleware
- API Suite Testing & QA Verification
- Technical Documentation, Presentation & Demo Setup

MILESTONES WORKFLOW

1. Concept & Design **2. Core Backend** **3. Weather Data** **4. AI Integration** **5. QA & Testing** **6. Demo & Docs**

Sequential execution ensuring weather data pipeline reliability prior to AI layer integration and QA verification.

COMPLETION CRITERIA

- Backend server runs successfully and connects to MongoDB without connection errors.
- Protected user routes require and correctly validate JWT authentication headers.
- Saved locations and cached weather telemetry operations execute reliably with low latency.
- Gemini AI endpoints process weather context and output relevant personalized advisories and alert summaries.
- Comprehensive API test suites pass successfully and are fully documented.
- Final project specification document, presentation slides, and live demonstration video are fully prepared.